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作业

3.1 Calculate the intrinsic carrier density n_i for silicon at $T = 50$ K and 350 K.

Ans. $9.6 \times 10^{-39}/\text{cm}^3$; $4.15 \times 10^{11}/\text{cm}^3$

$$\boxed{n_i = BT^{3/2} e^{-E_g/2kT}}$$

where B is a material-dependent parameter that is $7.3 \times 10^{15}\,\text{cm}^{-3}\text{K}^{-3/2}$ for silicon; T is the temperature in K; E_g, a parameter known as the **bandgap energy**, is 1.12 electron volt (eV) for silicon[2]; and k is Boltzmann's constant (8.62×10^{-5} eV/K).

$n_i = B \cdot T^{3/2} \cdot e^{-E_g/2kT}$ ，代入 $T = 50$K，350K，得到

$$n_i = \begin{cases} 9.254 \times 10^{-39}/\text{cm}^3 & @\ T = 50\text{K} \\ 4.128 \times 10^{11}/\text{cm}^3 & @\ T = 350\text{K} \end{cases}$$

3.3 For a silicon crystal doped with boron, what must N_A be if at $T = 300$ K the electron concentration drops below the intrinsic level by a factor of 10^6?

Ans. $N_A = 1.5 \times 10^{16}/\text{cm}^3$

$$\boxed{p_p \simeq N_A \qquad p_p n_p = n_i^2 \qquad n_i \simeq 1.5 \times 10^{10}/\text{cm}^3}$$

$p = N_A$, $n = \dfrac{n_i^2}{p}$ $\qquad n = \dfrac{n_i}{10^6}$ $\qquad \Rightarrow p = n_i \cdot 10^6 = 1.5 \times 10^{16} /\text{cm}^3$

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$\boxed{v_{n\text{-drift}} = -\mu_n E}$ $\boxed{J = J_p + J_n = q(p\mu_p + n\mu_n)E}$

3.4 A uniform bar of n-type silicon of 2 µm length has a voltage of 1 V applied across it. If $N_D = 10^{16}/\text{cm}^3$ and $\mu_n = 1350\ \text{cm}^2/\text{V}\cdot\text{s}$, find (a) the electron drift velocity, (b) the time it takes an electron to cross the 2-µm length, (c) the drift-current density, and (d) the drift current in the case the silicon bar has a cross sectional area of 0.25 μm^2.

Ans. 6.75×10^6 cm/s; 30 ps; $1.08 \times 10^4\ \text{A/cm}^2$; 27 µA

(a) $E = \frac{1V}{2\mu m}$ $\quad v = -\mu_n E = -1350\ \text{cm}^2/\text{V}\cdot\text{s} \cdot \frac{1V}{2\mu m} = -6.75 \cdot 10^6\ \text{cm/s}$

(b) $t = \frac{2\mu m}{v} = 29.63\ \text{ps}$

(c) $J = q n \mu_n E = q \cdot N_D \cdot \mu_n \cdot E = 1.081 \cdot 10^4\ \text{A/cm}^2$

(d) $I = J \cdot 0.25\mu m^2 = 27.037\ \mu A$

3.5 The linear electron-concentration profile shown in Fig. E3.5 has been established in a piece of silicon. If $n_0 = 10^{17}/\text{cm}^3$ and $W = 0.5\ \mu\text{m}$, find the electron-current density in microamperes per micron squared ($\mu\text{A}/\mu\text{m}^2$). If a diffusion current of 1 mA is required, what must the cross-sectional area (in a direction perpendicular to the page) be? Recall that $D_n = 35\ \text{cm}^2/\text{s}$.

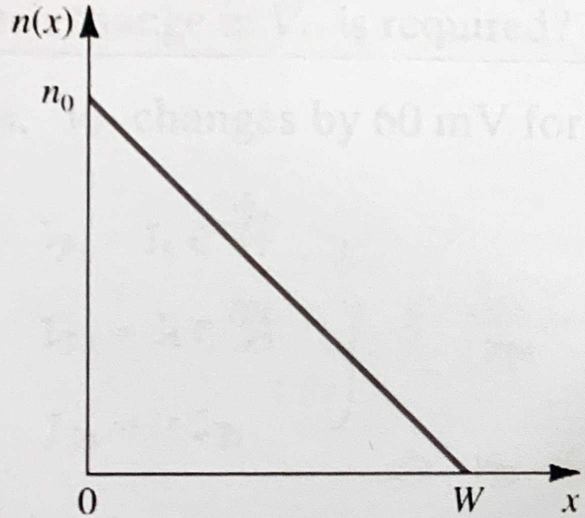

Figure E3.5

$$\boxed{J_{tot} = q\left(D_n\frac{dn}{dx} - D_p\frac{dp}{dx}\right)}$$

$$\frac{dn}{dx} = \frac{n_0}{W}$$

$$J = q \cdot D_n \cdot \frac{dn}{dx} = 112.152\ \mu\text{A}/\mu\text{m}^2$$

$$A = \frac{I}{J} = \frac{1\text{mA}}{112.152\ \mu\text{A}/\mu\text{m}^2} = 8.916\ \mu\text{m}^2$$

Ans. 112 $\mu\text{A}/\mu\text{m}^2$; 9 μm^2

A diode operates in the forward bias region with a typical current level [i.e., $I_D \approx I_S \exp(V_D/V_T)$]. Suppose we wish to increase the current by a factor of 10. How much change in V_D is required?

Ans. V_D changes by 60 mV for a decade (tenfold) change in I_D

$$I_{D1} = I_s e^{\frac{V_{D1}}{V_T}}$$

$$I_{D2} = I_s e^{\frac{V_{D2}}{V_T}}$$

$$I_{D2} = 10 I_{D1}$$

$$\Rightarrow \frac{I_{D2}}{I_{D1}} = e^{\frac{V_{D2}-V_{D1}}{V_T}} = 10$$

$$\Rightarrow V_{D2} - V_{D1} = \ln(10) \cdot 26\,mV$$

$$= 59.867\,mV$$

$$\approx 60\,mV$$